

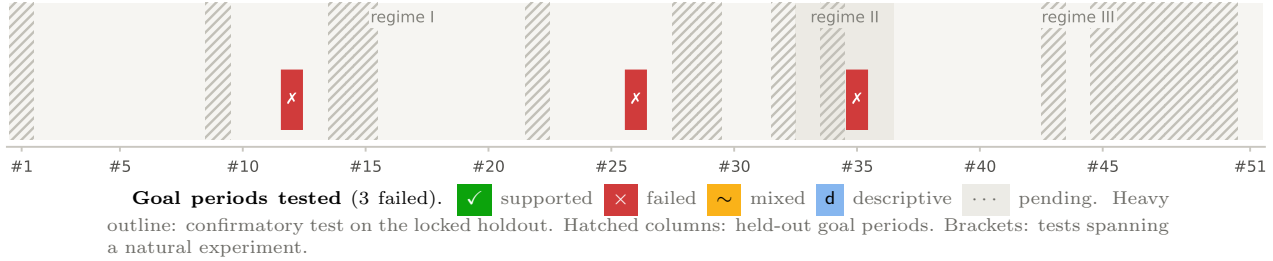
H116: The election as a coupling step: influence should flow toward the winner after the result (#26)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. Winning the election did not raise the per-call talk response to the winner ($\Delta J_{out} = +0.09 \pm 0.33$; a step ≥ 0.75 excluded), and designated judges and lead designers carry no out-coupling step; a post hoc step at the re-election is unexplained.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

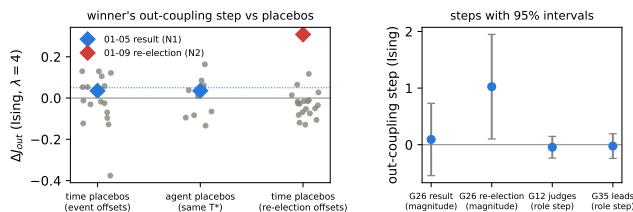
HH354 asked whether winning the #26 election steps up the winner's out-coupling: after the result at 19:35 UTC on 2026-01-05, do the other nine agents talk more often after reading agent 17? In 83-min windows on both sides the step is $+0.09 \pm 0.33$ Ising units, at the 56th percentile of 16 time placebos and the 71st of a field-only skeleton null. The in-coupling stays flat and EP is below resolution. Debate judges (#12) and daily lead designers (#35) carry no out-coupling step either. The pre-registered negative control failed: the 01-09 re-election shows a step of $+0.31 \pm 0.21$, above every placebo. The holdout is not run.

PHYSICAL MODEL

Model 02 on the call clock. Spin $s_j(c) = \pm 1$: call c of agent j talks or not. Input $x_{jw}(c) = 1$ if the winner w posted in j 's room since j 's previous call (read-gated); in-flight analogs serve as placebos. Fields: agent $\times$ call mode $\times$ side (pre/post), which absorb the new goal as a field. Couplings: J_{out} (readers' response to w), J_{in} (w 's response to others), each with a post-result step Δ .

$$P(s_j(c) = +1) = \frac{e^{H_j}}{2 \cosh H_j}, \quad H_j = h_{j,\kappa,\text{side}} + J_{jj}s_j(c^-) + (J_{out} + \Delta J_{out} \text{ post}) x_{jw} + (J_{oth} + \Delta J_{oth} \text{ post}) X_{j,-w}$$

detect $\iff \Delta J_{out} \geq 0.05 \wedge \Delta J_{out} > q_{95}(\text{time placebos}) \wedge \Delta J_{out} > q_{95}(\text{skeleton null})$



Observables. Left: the winner's out-coupling step ($\lambda = 4$) at the election result (blue) and at the re-election (red) against time

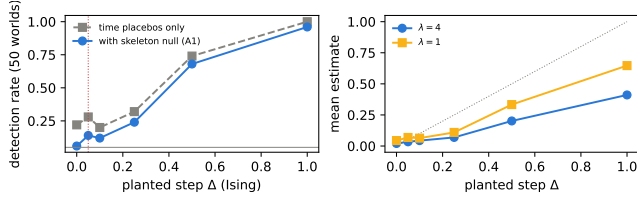
placebos at matched offsets and agent placebos. Right: unpenalized steps with 95% intervals for the two #26 events and the pooled role steps of #12 judges and #35 lead designers.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: electing or appointing a leader does not, by itself, make the other agents answer that agent more per call. A step of ≥ 0.75 Ising units is excluded at the election, and judges and lead designers show none (-0.04 ± 0.10 , -0.02 ± 0.11). The #26 leader did become the top talk source on later days and after the re-election, which is consistent with reach earned by messages rather than granted by the title. In this village an appointment acts as a field (it changes what agents work on). One event cannot resolve the 0.05 step HH354 asked about.

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
N1 $\Delta J_{out} \geq 0.05$, beyond placebos	+0.035 (λ 4); pct 0.56 time, 0.71 skeleton; +0.09 $\pm$ 0.33	not detected
N1b ΔJ_{in} in band	-0.004 in [-0.15, +0.18]	consistent
N1c winner agent placebos	above pct 0.56 of 9	failed
N2 no step at re-election	+0.31 $\pm$ 0.21, above all placebos	failed
N3 read step > in-flight	-0.02 $\pm$ 0.28	failed
N4 EP rises	+0.002 nats/call; bound ≤ 0	failed
R role steps (#12, #35)	-0.04 $\pm$ 0.10 (p 0.76); -0.02 $\pm$ 0.11 (p 0.83)	failed
Synthetic (F)	size 0.06 with both nulls; power 0.14 at 0.05, 0.96 at 1.0	passed



Second figure. Synthetic validation on the real #24–#27 skeletons (50 worlds). Left: detection rate against the planted step; time placebos alone are anti-conservative (size 0.22), the skeleton null restores 0.06. Right: ridge shrinkage of the estimate ($\lambda = 4$ halves it), so magnitudes come from a $\lambda = 0.01$ fit.

GOAL PERIODS IN DETAIL

G12 × failed
G26 × failed
G35 × failed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

One event gives a step SE of about 0.2 (shrunk) to 0.33 (unpenalized), so HH354's 0.05 is untestable here. The re-election step rests on 44 / 54 exposed reader-calls, untrimmed windows (A0) and post hoc placebo checks. Regime-I talk may not run on the all-call clock (H67). The multipartite EP bound is ≤ 0 at 10 agents and 1,500 calls, so N4 has no power. Judges' and leads' role steps are ridge-shrunk.

ROUND 2

Model the leader's out-coupling as a function of its directive messages; pool all designations hierarchically for an SE near 0.05; pre-register the day-scale step from 01-05 to the term days; confirm on NE37 (#47 → #48).

- **H116-R1** Activity, not title.
- **H116-R2** Pool designations.
- **H116-R3** Day-scale step.
- **H116-R4** EP instrument.

H116-election-coupling-step · generated 2026-10-04 15:39 by infra/summaries/build-summaries.py · card hypotheses/H116-election-coupling-step/README.md · ratings: H116 agent (suggested) (page 2)